The Biosphere and the Discount Rate:

The Host with No Outside

Macharia Barii

October 2026

Abstract

Every previous predatory relation had an outside: new lands, new peoples, new resources. When the host was exhausted, the predator moved on. The biosphere has no outside. This paper argues that this is a new situation in the history of civilization, and that it changes the logic of conservation. When a host is unique and non-substitutable, the conservation principle fails: the predator cannot afford to be patient, because patience means losing to rivals who are impatient. The result is a race to consume the host before someone else does. The only correction is to lower the discount rate—to lengthen the time horizon of the order as a whole. The paper formalizes the conservation principle, derives the failure of conservation under uniqueness, and identifies the discount rate as the decisive political variable. It concludes that the question of the biosphere is not whether it matters—it does—but whether the political system can be made patient enough to conserve it.

Keywords: biosphere, discount rate, conservation, time horizon, non-substitutability, ecological accounting, intergenerational trustees

Contents

1 The Conservation Principle **2**

2 The Outside **2**

3 The Host with No Outside **2**

4 The Discount Rate as a Political Variable **3**

5 The Race to the Bottom **3**

6 The Correction **3**

7 Conclusion **4**

1 The Conservation Principle

The ledger framework derives a conservation principle from the patience of the party that controls a host. A party with a long time horizon (a low discount rate r) conserves its hosts, because the future value of the host exceeds the present value of consuming it. A party with a short time horizon (a high discount rate r) consumes its hosts, because the present value of consumption exceeds the discounted future value of the host.

Proposition 1.1 (The Conservation Principle)**.** *The optimal steady-state conservation of a host is decreasing in the discount rate r of the party that controls it. A patient party conserves; an impatient party consumes.*

This principle explains a great deal of history. A dynasty that expects to last a thousand years invests in infrastructure. A junta facing the next coup loots the treasury. A democracy facing the next election under-invests in everything that requires long-term commitment—infrastructure, education, ecological conservation. The difference is not virtue; it is the discount rate.

2 The Outside

Every previous predatory relation had an outside. When the host was exhausted, the predator moved on. The Roman Empire exhausted the soils of Italy and moved to the provinces. The colonial powers exhausted one colony and moved to the next. The whaling fleets exhausted one fishery and moved to another. The outside was the escape valve that made predation sustainable: the predator could always find a new host.

The outside had a second function. It made patience rational. A predator who conserved a host was conserving it for a rival. The predator who consumed the host now, before a rival could, captured the value. The outside therefore pushed the discount rate up: with a new host always available, the value of conserving the current host was low.

3 The Host with No Outside

The biosphere has no outside. This is a new situation in the history of civilization.

Proposition 3.1 (The Failure of Conservation Under Uniqueness)**.** *When the host is unique and non-substitutable, the conservation principle fails. The predator cannot afford to be patient, because patience means losing to rivals who are impatient. The result is a race to consume the host before someone else does.*

The logic is the same as the tragedy of the commons, but with a difference. In the classic commons, the failure is a coordination failure: each actor would prefer that all conserve, but each has an incentive to defect. In the biosphere, the failure is deeper: there is no outside to which the predator can move, and no substitute for the host. The race to consume is not a failure of coordination; it is the rational response to a unique, non-substitutable host.

Every previous civilization collapsed into its own locality. This one has nowhere to collapse to. The question is not whether the biosphere matters—it does—but whether the political system can be made patient enough to conserve it.

4 The Discount Rate as a Political Variable

The framework says that the answer is determined by the discount rate, and that the discount rate is a political variable.

Proposition 4.1 (The Discount Rate Is Political)**.** *The discount rate of the coalition that sustains government is not a fact of nature. It is a parameter set by institutions: constitutional rules, long-horizon funds, intergenerational trustees, and ecological accounting all lower the discount rate by lengthening the time horizon of the order.*

This is the key insight. The discount rate is not given; it is chosen. A coalition that faces only the next election has a high discount rate. A coalition that is constrained by a constitution that outlives it has a lower one. A coalition that is accountable to an intergenerational trustee has a lower one still. The institutions that lengthen the horizon are not moral gestures; they are attempts to raise the patience of the political system, because patience is the only thing that can conserve a host that has no substitutes.

Corollary 4.1. *Constitutional rules, independent central banks, long-horizon sovereign funds, intergenerational trustees, and ecological accounting are all attempts to lower the discount rate by lowering the dependence of policy on the electoral or factional cycle.*

5 The Race to the Bottom

The failure of conservation under uniqueness has a further consequence: it produces a race to the bottom among predators. If the host is unique, and if patience means losing to rivals, then each predator has an incentive to consume faster than its rivals. The result is a competitive acceleration of consumption, in which the discount rate of the whole rises as each actor tries to out-consume the others.

This is why the biosphere is being consumed faster than any previous host. It is not that the actors are more wicked than their predecessors. It is that the host is unique, and the uniqueness removes the incentive for patience. The race to the bottom is the structural consequence of the host with no outside.

6 The Correction

The only correction is to lower the discount rate—to lengthen the time horizon of the order as a whole. This is what constitutional rules, intergenerational trustees, and ecological accounting attempt to do. They are not moral gestures. They are attempts to raise the patience of the political system, because patience is the only thing that can conserve a host that has no substitutes.

The correction has three components.

First, lengthen the horizon of the coalition. The coalition that sustains government must be made to care about the next generation, not just the next election. This requires institutions that outlive the incumbents: constitutional rules, independent trustees, long-horizon funds.

Second, make the host legible. The biosphere is invisible to the ledger because it is not priced. Ecological accounting makes the host legible by pricing the depletion of the host into the ledger. What is not measured is not conserved.

Third, remove the race. The race to the bottom is a coordination failure among predators. It can be corrected only by a coordination mechanism that removes the incentive to defect: international agreements, enforceable limits, and institutions that make patience rational.

7 Conclusion

Every previous predatory relation had an outside. The biosphere has no outside. This is a new situation in the history of civilization, and it changes the logic of conservation.

When the host is unique and non-substitutable, the conservation principle fails. The predator cannot afford to be patient, because patience means losing to rivals who are impatient. The result is a race to consume the host before someone else does.

The only correction is to lower the discount rate—to lengthen the time horizon of the order as a whole. This is what constitutional rules, intergenerational trustees, and ecological accounting attempt to do. They are not moral gestures. They are attempts to raise the patience of the political system, because patience is the only thing that can conserve a host that has no substitutes.

The framework does not answer the question of whether the political system can be made patient enough. It says that the answer is determined by the discount rate, and that the discount rate is a political variable. The question is not whether the biosphere matters—it does—but whether the forces contesting the ledger can lower the discount rate before the host is consumed.

The question is always the same: whose ledger is this, and for whom? For the biosphere, the answer is: the present's, and at the future's expense. The correction is to make the answer broader—to include the future in the ledger, and to make the present pay for what it consumes.

The rest is history—or its absence.